AI4Science Meeting 2

Literature Review & Proposed Methods

Common Practices

- Retrieval: Match with images sharing the most similar embedded features

1) hypothesis: pictures in similar positions share similar features
2) Pros: easily match granular labels across multiple geocells
3) Cons: heavily reliant on the distribution of reference data

- Classification: establish geocells as classifier labels

1) Pros: learn features relating to more ‘general’ features
2) Cons: unable to conduct further prediction within each cell; tradeoff between cell accuracy and training difficulty

- Generative: diffusion model-based prediction
- Hybrid: a mixed methodology across retrieval/classification/regression…

References

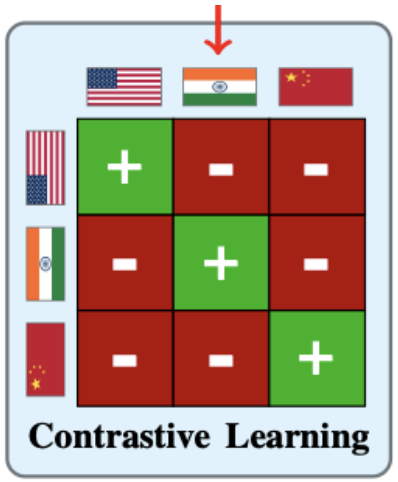
- Around the World in 80 Timesteps: A Generative Approach to Global Visual Geolocation (Dec 2024) – PLONK
- PIGEON: PREDICTING IMAGE GEOLOCATIONS (May 2024) – PIGEON
- OpenStreetView-5M: The Many Roads to Global Visual Geolocation (Apr 2024)
- GT-Loc: Unifying When and Where in Images Through a Joint Embedding Space (Jul 2025)
- GeoToken: Hierarchical Geolocalization of Images via Next Token Prediction (Nov 2025)
- Img2Loc: Revisiting Image Geolocalization using Multi-modality Foundation Models and Image-based Retrieval-Augmented Generation (Mar 2024)

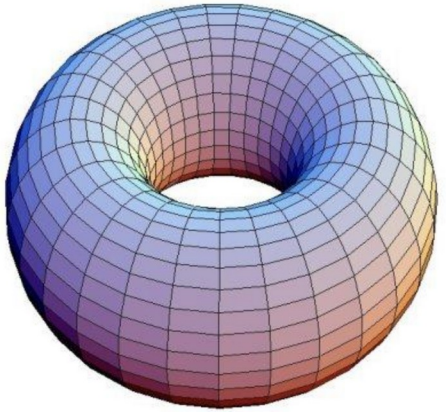
Data Augmentation & Handling

- Incorporate additional semantic labels, encourage encoder to extract more geographical features
- Contrastive learning: increase distance of embedded features across different regions
- Representing time: (month, hour) pair, reformulated in torus

- **Location**: “A photo I took in the region of Gauteng in South Africa."
- **Climate**: “This location has a temperate oceanic climate."
- **Compass direction**: “This photo is facing north."
- **Season (month)**: “This photo was taken in December."
- **Traffic**: “In this location, people drive on the left side of the road."

$$\mathcal{L}_n = \mathcal{L}_{n,\text{loc}} + \alpha\mathcal{L}_{n,\text{climate}} + \beta\mathcal{L}_{n,\text{month}} + \gamma\mathcal{L}_{n,\text{reg}}$$

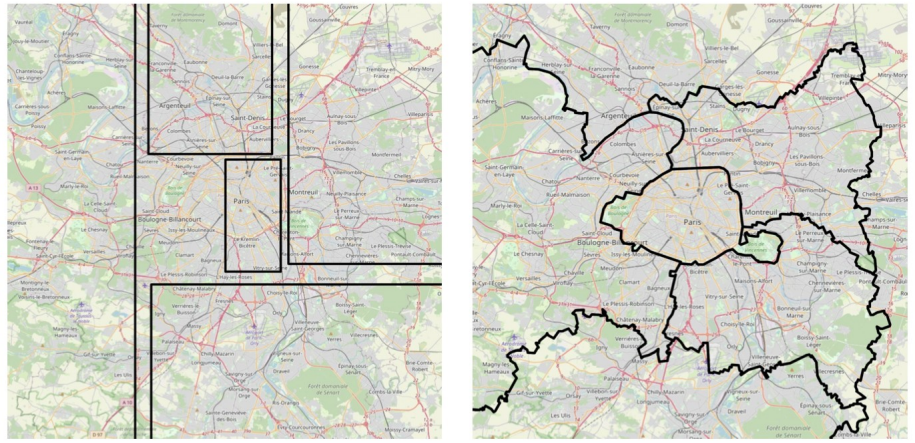




Formulating geocells

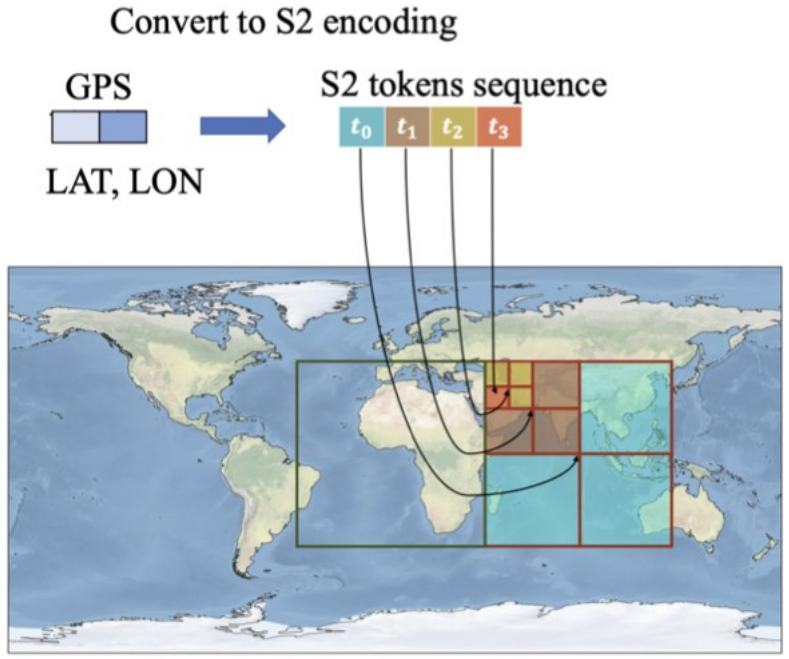
- Naïve approach: rectangular cells
- Semantic approach: using administrative information
- Adapt size of each geocell to ensure minimal sample number in each

- Other predefined encoding (S2)



(a) With naive, rectangular geocells.

(b) With our semantic geocells.



Loss Function

- Smoothened labels: penalize distance between cells
- Consider joint loss regarding likelihood of different predictions

$$y_{n,i} = \exp\left(-\frac{\mathrm{Hav}(\mathbf{g}_i, \mathbf{x}_n) - \mathrm{Hav}(\mathbf{g}_n, \mathbf{x}_n)}{\tau}\right)$$

$$\mathcal{L}_n = -\sum_{g_i \in G} \log\left(p_{n,i}\right) \cdot \exp\left(-\frac{\mathrm{Hav}(\mathbf{g}_i, \mathbf{x}_n) - \mathrm{Hav}(\mathbf{g}_n, \mathbf{x}_n)}{\tau}\right)$$

Methodology

$$c^* = \arg\min_{c \in C_g} \left\| f(x) - \frac{1}{|c|} \sum_{l \in c} f(l) \right\|_2$$

- Geocell classification + retrieval-based refinement (clustering)

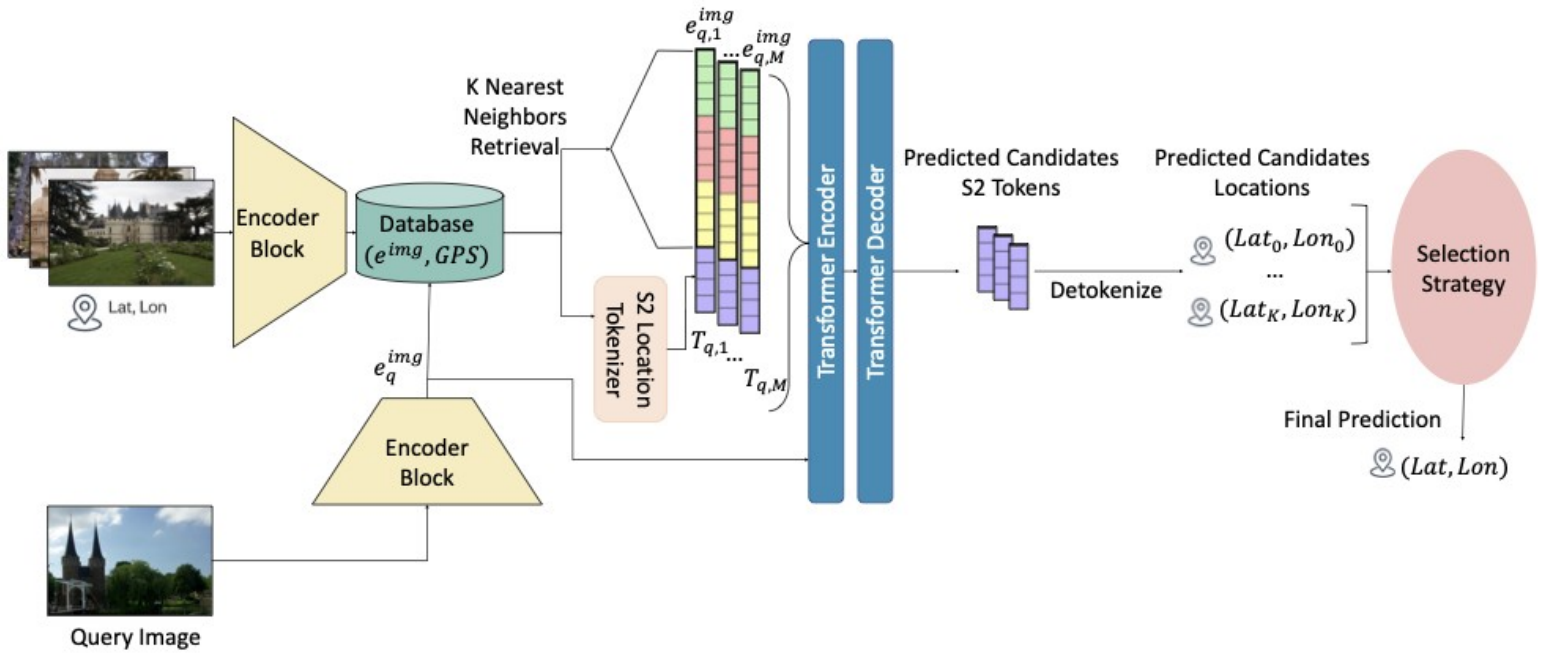
P(label) = p(geocell)*softmax(distance(embedding))

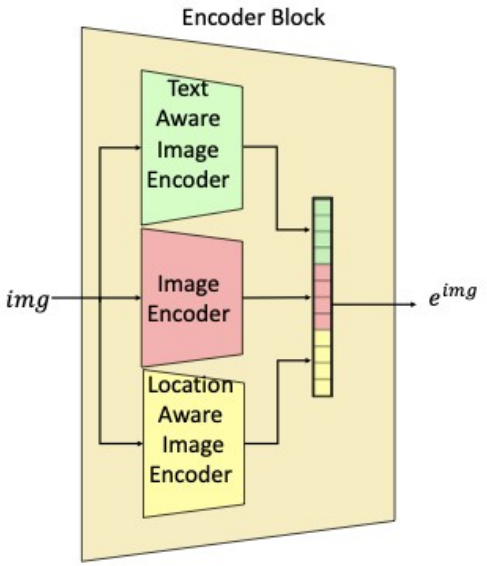
- Direct Regression: features-(longitude, latitude)
- Geocell classification + relative position regression

R^d-[−1,1] normalization: centroid of datasets = (0,0)

- Use LLM as end-point classifier

Transformer





$$\mathcal{L} = \frac{1}{\sum_t w_t} \sum_{t=0}^{20} w_t \, \mathrm{CE}(\hat{y}_t, y_t), \quad \text{where } w_t = 2.0 - \frac{t}{20}$$

Weights early layers

Decoding Strategy:

$$\mathbf{p}_t = \mathrm{softmax}\left(\frac{\boldsymbol{\ell}_t}{T}\right).$$

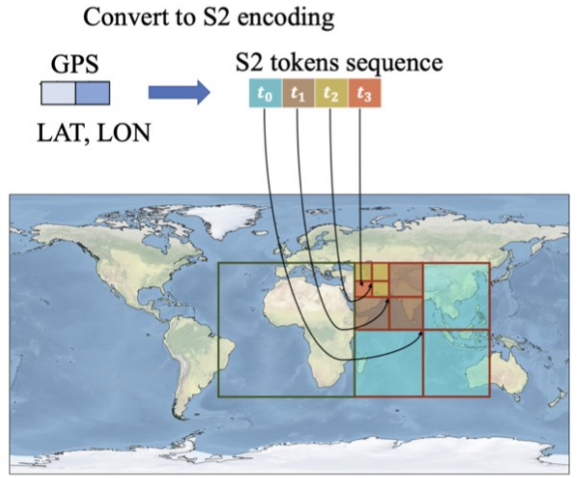
Sampling with Temperature

/

$$\mathrm{score}(\tau \oplus k) = \mathrm{score}(\tau) + \log p(k \mid \tau),$$

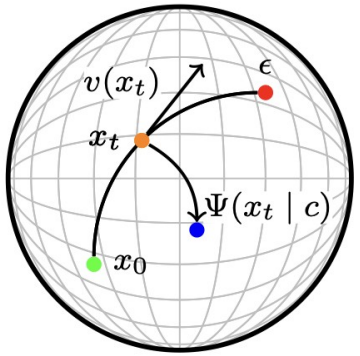
Beam Search

Candidate selection: log-probability/ Reward Model reranking/ Similarity -based



Diffusion Model: PLONK YFCC

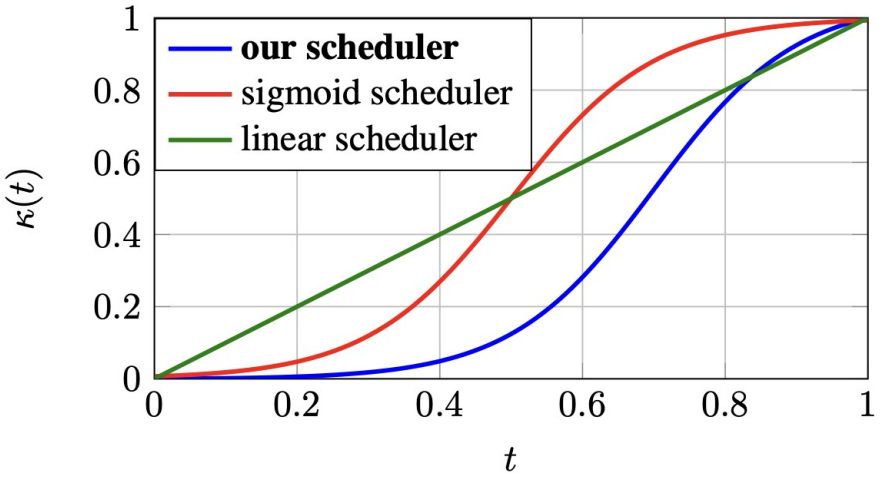
- Diffusion problems in Euclidean & Rieman Space (a spherical surface)
- The parameter of scheduler implies concentrated attention on more general/ localized feature
- Architecture: 6*(MLP+GeLU)+residual block



- x_0: true location
- ϵ: sampled noise
- x_t: noisy location
- $\psi(x_t \mid c)$: prediction
- $\rightarrow v(x_t)$: velocity field

Diffusion
$x_t = \sqrt{1-\kappa(t)}x_0 + \sqrt{\kappa(t)}\epsilon$ $\mathcal{L}_{\text{D}} = \|\psi(x_t \mid c) - \epsilon\|^2$
Flow Matching
$x_t = (1-\kappa(t))x_0 + \kappa(t)\epsilon$ $\mathcal{L}_{\text{FM}} = \|\psi(x_t \mid c) - v(x_t)\|^2$
Riemannian Flow Matching
$x_t = \exp_{x_0}\left(\kappa(t)\log_{x_0}(\epsilon)\right)$ $\mathcal{L}_{\text{RFM}} = \|\psi(x_t \mid c) - v(x_t)\|^2_{x_t}$

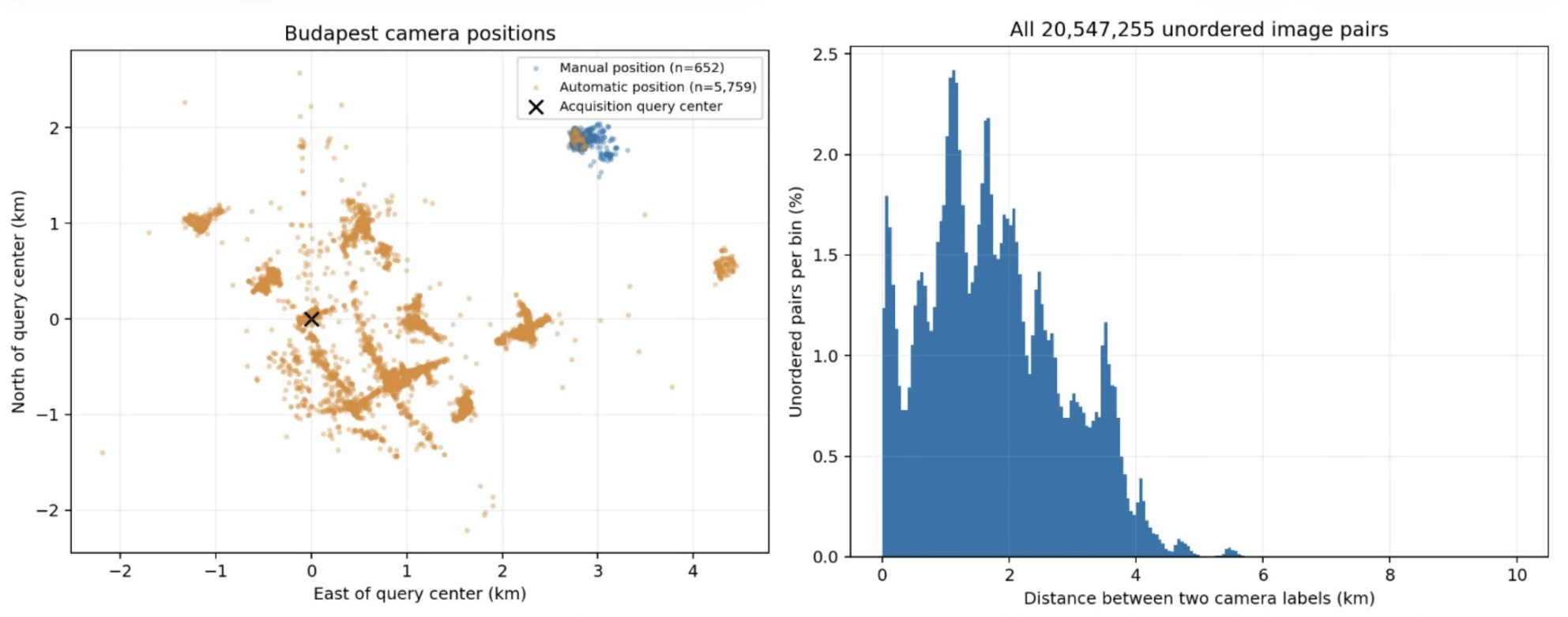
$\kappa(t)$: noise scheduler



Performances: 9103 pictures in both datasets

Dataset	Model	N	Mean km	Median km	P90 km	≤1 km %	≤25 km %	≤200 km %	≤750 km %	≤2500 km %
smapshot	geoclip	9103	1629.98	206.43	6709.22	0.60	10.24	49.36	75.78	85.96
smapshot	osv5m	9103	2428.49	496.95	8682.26	0.00	2.82	26.61	58.60	76.38
smapshot	gtloc	9103	1322.64	183.58	5911.65	0.66	11.46	53.36	80.31	87.41
smapshot	plonk_osv	9103	826.39	269.86	1542.44	0.00	4.04	41.06	83.68	93.52
smapshot	plonk_yfcc	9103	942.23	108.55	1690.32	0.22	14.43	68.56	87.78	90.17
urbanhistory4d	geoclip	9103	1118.17	448.52	1849.21	22.28	27.57	33.60	68.34	92.99
urbanhistory4d	osv5m	9103	2562.75	1415.17	7925.94	0.02	0.74	5.21	29.57	74.85
urbanhistory4d	gtloc	9103	1313.67	502.70	2103.59	19.47	23.48	29.61	64.17	91.37
urbanhistory4d	plonk_osv	9103	1157.91	448.67	1713.32	0.01	1.96	19.48	66.35	93.24
urbanhistory4d	plonk_yfcc	9103	490.54	3.10	964.42	29.89	50.92	59.23	86.73	97.09

Data Distribution



- Smapshot: 355,250 publicly available data with valid labels
- 4dbrowser: 29,035 publicly available data with valid labels

Summary on Literature

- Majority of literature adopt classical retrieval/ classification methods, with moderate performance
- Diverse hybrid methods are observed, showing promising results

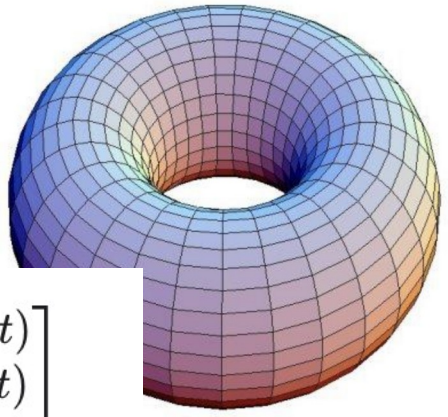
Research Questions

1) How to encode historical information (time of picture, historical geographical information, etc.) to facilitate prediction

2) How to capture the localizability of pictures and reliability of each prediction

Historical Information Representation

- Not to predict timestamp, but extract information to assist predicting geolocation



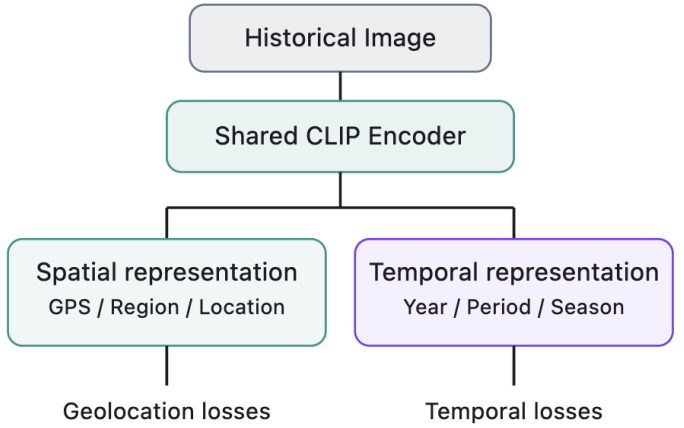
$$h(t) = \begin{bmatrix} a\cos(2\pi t) \\ a\sin(2\pi t) \\ bt \end{bmatrix}$$

- Spiral representation or soft targets

$$w_{ij} = \exp\left(-\frac{|t_i - t_j|}{\tau_y}\right)$$

- Better comparison: same picture taken at different time

- Availability in dataset remains unchecked



Localizability: intrinsic feature

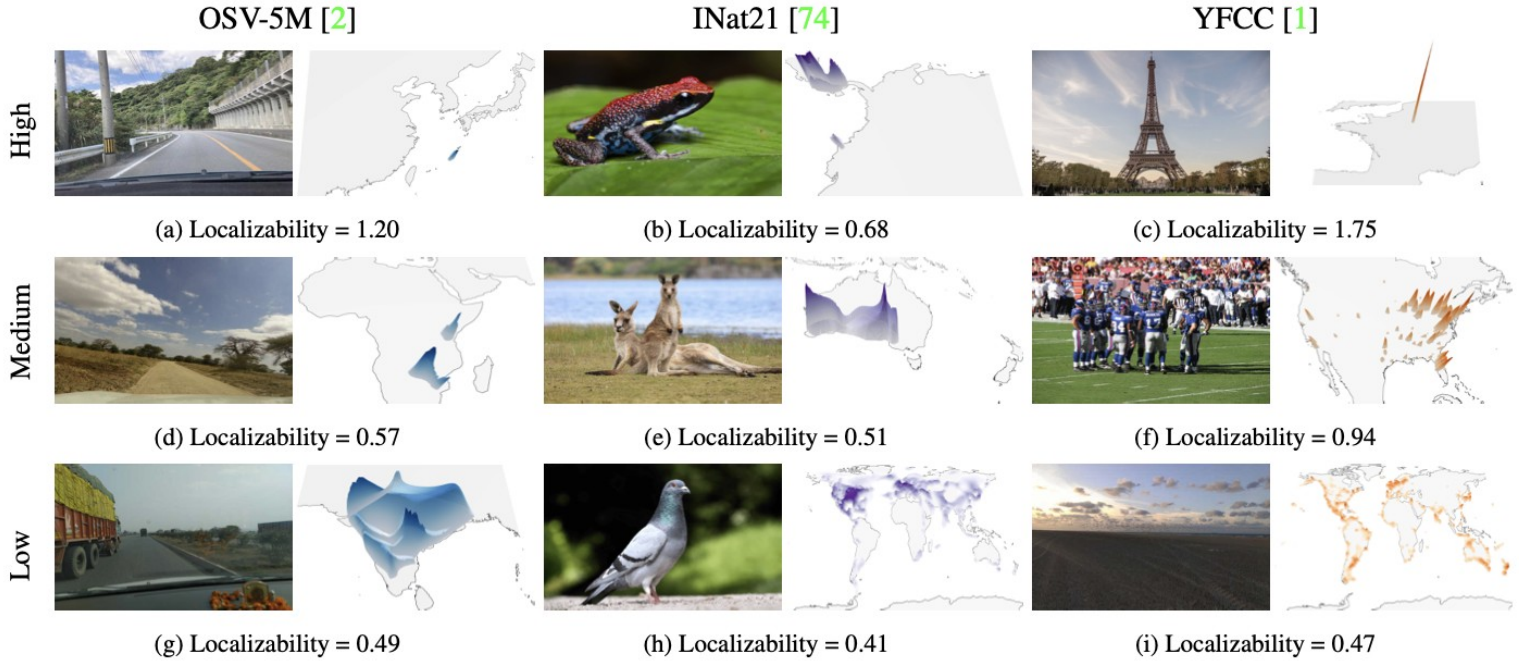


Figure 6. **Estimating Localizability.** We use the entropy of the predicted distribution as a proxy for the localizability of images. For each dataset, we present examples of high, medium, and low localizability, which correlate well with human perception.

- Given historical background and geological information, some pictures are empirically easier to localize than others
- Such features, however, are easily masked by the performance of individual models and are hard to generalize

Localizability: Model Outputs

- Infer reliability from model outputs: distinguish confidence from accuracy

$$A_\psi\left(z_I,\, \mathcal{C}(I),\, p,\, s_{\text{rank}},\, \hat{x}\right) \longrightarrow \begin{bmatrix} P(e \leq 100\text{m}) \\ P(e \leq 1\text{km}) \\ P(e \leq 5\text{km}) \\ P(e \leq 25\text{km}) \end{bmatrix}$$

- Zi ~ embedded features of each photo
- C(I) ~ candidate (longitude, latitude) pairs
- P ~ original likelihood of each candidate
- Srank ~ rank assigned to each candidate by ranker
- X ~ final output of model

- E: error of prediction

Cross-Validation (e.g. 5-fold):

1) Divide the training set into 5 folds, each time use 4 folds to train the feature extraction model and use the last 1 to attain inputs for accuracy prediction model, repeat 5 times
2) Use the attained inputs to train the accuracy model

Proposed Structure

- Represent the GPS location in Swiss coordinates
- Use GeoCLIP as backbone (with CLIP frozen) to extract features in images & (longitude, latitude) pairs
- Add auxiliary features (regions/ time) to guide the attention of encoders
- Grid geocells with regards to administrative data & data availability
- Perform regression within cells with regards to clusters to attain finer prediction
- Preserve multiple likely predictions, design & train rankers to identify most likely outputs
- Use cross-validation to train prediction model on accuracy of outputs